

HW10

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Exercise 10.1

设二维离散随机变量 (X, Y) 的可能取值为

$$(0, 0), \quad (-1, 1), \quad (-1, 2), \quad (1, 0),$$

且取这些值的概率依次为 $1/6, 1/3, 1/12, 5/12$, 试求 X 与 Y 各自的边际分布列.

Solution.

$$P(X = t, Y < \infty) = \begin{cases} \frac{5}{12}, & t = 1 \\ \frac{1}{6}, & t = 0 \\ \frac{1}{3} + \frac{1}{12} = \frac{5}{12}, & t = -1 \\ 0, & \text{其它} \end{cases}$$

$$P(X < \infty, Y = t) = \begin{cases} \frac{1}{12}, & t = 2 \\ \frac{1}{3}, & t = 1 \\ \frac{1}{6} + \frac{5}{12} = \frac{7}{12}, & t = 0 \\ 0, & \text{其它} \end{cases}$$

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Exercise 10.2

设二维随机变量 (X, Y) 的联合分布函数为

$$F(x, y) = \begin{cases} 1 - e^{-\lambda_1 x} - e^{-\lambda_2 y} + e^{-\lambda_1 x - \lambda_2 y - \lambda_{12} \max\{x, y\}}, & x > 0, \quad y > 0, \\ 0, & \text{其他.} \end{cases}$$

试求 X 和 Y 各自的边际分布函数.

Solution.

$$F_X(x) = F(x, \infty) = \begin{cases} 1 - e^{-\lambda_1 x}, & x > 0 \\ 0, & \text{其它} \end{cases}$$

X	-1	0	1
P	$\frac{5}{12}$	$\frac{1}{6}$	$\frac{5}{12}$

$$F_Y(x) = F(\infty, y) = \begin{cases} 1 - e^{-\lambda_2 y}, & y > 0 \\ 0, & \text{其它} \end{cases}$$

Y	0	1	2
P	$\frac{7}{12}$	$\frac{1}{3}$	$\frac{1}{12}$

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Exercise 10.3

求以下给出的 (X, Y) 的联合密度函数的边际密度函数 $p_X(x)$ 和 $p_Y(y)$:

1.

$$p(x, y) = \begin{cases} e^{-y}, & 0 < x < y, \\ 0, & \text{其他;} \end{cases}$$

2.

$$p(x, y) = \begin{cases} \frac{5}{4}(x^2 + y), & 0 < y < 1 - x^2, \\ 0, & \text{其他;} \end{cases}$$

3.

$$p(x, y) = \begin{cases} \frac{1}{x}, & 0 < y < x < 1, \\ 0, & \text{其他.} \end{cases}$$

Solution.

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y p(s, t) dt ds = \int_{-\infty}^y \int_{-\infty}^x p(s, t) ds dt$$

$$\begin{aligned}
 p_X(x) &= \frac{d}{dx} \lim_{y \rightarrow \infty} \int_{-\infty}^x \int_{-\infty}^y p(s, t) dt ds \\
 &= \frac{d}{dx} \int_{-\infty}^x \int_{-\infty}^{\infty} p(s, t) dt ds \\
 &= \int_{\mathbb{R}} p(x, y) dy
 \end{aligned}$$

类似地,

$$p_Y(y) = \int_{\mathbb{R}} p(x, y) dx$$

1.

$$p_X(x) = \int_{\mathbb{R}} \chi_{\{y>x\}} \chi_{\{x>0\}} e^{-y} dy = \begin{cases} \int_x^{\infty} e^{-y} dy = e^{-x}, & x > 0 \\ \int_{\mathbb{R}} 0 dy = 0, & x \leq 0 \end{cases}$$

$$p_X(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$p_Y(y) = \int_{\mathbb{R}} \chi_{\{x<y\}} \chi_{\{x>0\}} e^{-y} dx = \begin{cases} \int_0^y e^{-y} dx = ye^{-y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$p_Y(y) = \begin{cases} ye^{-y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

2.

$$\begin{aligned}
 p_X(x) &= \int_{\mathbb{R}} \chi_{\{y>0\}} \chi_{\{1-x^2>y\}} \frac{5}{4} (x^2 + y) dy \\
 &= \chi_{\{1-x^2>0\}} \int_0^{1-x^2} \frac{5}{4} (x^2 + y) dy \\
 &= \chi_{\{1-x^2>0\}} \frac{5}{4} \left((x^2)(1-x^2) + \frac{1}{2} (1-x^2)^2 \right) \\
 &= \chi_{\{1-x^2>0\}} \frac{5}{4} (1-x^2) \left(\frac{1}{2} + \frac{1}{2} x^2 \right) = \chi_{\{1-x^2>0\}} \frac{5}{8} (1-x^4)
 \end{aligned}$$

因此

$$p_X(x) = \begin{cases} \frac{5}{8} (1-x^4), & -1 < x < 1 \\ 0, & \text{其它} \end{cases}$$

$$\begin{aligned}
p_Y(y) &= \int_{\mathbb{R}} \chi_{\{1>y>0\}} \chi_{\{x^2<1-y\}} \frac{5}{4} (x^2 + y) \, dx \\
&= \chi_{\{1>y>0\}} \int_{-\sqrt{1-y}}^{\sqrt{1-y}} \frac{5}{4} (x^2 + y) \, dx \\
&= \chi_{\{1>y>0\}} \frac{5}{4} \left(\frac{2}{3} (1-y)^{\frac{3}{2}} + 2y\sqrt{1-y} \right) \\
&= \chi_{\{1>y>0\}} \frac{5}{6} \sqrt{1-y} (1+2y)
\end{aligned} \tag{10.1}$$

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$$p_Y(y) = \begin{cases} \frac{5}{6} \sqrt{1-y} (1+2y), & 0 < y < 1 \\ 0, & \text{其它} \end{cases}$$

3.

$$\begin{aligned}
p_X(x) &= \int_{\mathbb{R}} \frac{1}{x} \chi_{\{0<y<x<1\}} \, dy \\
&= \chi_{\{0<x<1\}} \int_0^x \frac{1}{x} \, dy \\
&= \chi_{\{0<x<1\}}
\end{aligned} \tag{10.2}$$

因此

$$\begin{aligned}
p_X(x) &= \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{其它} \end{cases} \\
p_Y(y) &= \int_{\mathbb{R}} \frac{1}{x} \chi_{\{0<y<x<1\}} \, dx \\
&= \chi_{\{0<y<1\}} \int_y^1 \frac{1}{x} \, dx \\
&= \chi_{\{0<y<1\}} \ln \frac{1}{y}
\end{aligned} \tag{10.3}$$

因此

$$p_Y(y) = \begin{cases} \ln \frac{1}{y}, & 0 < y < 1 \\ 0, & \text{其它} \end{cases} \tag{10.4}$$

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Exercise 10.4

试验验证：以下给出的两个不同的联合密度函数，它们有相同的边际密度函数.

$$p(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, \quad 0 \leq y \leq 1, \\ 0, & \text{其他;} \end{cases}$$

$$g(x, y) = \begin{cases} (0.5 + x)(0.5 + y), & 0 \leq x \leq 1, \quad 0 \leq y \leq 1, \\ 0, & \text{其他.} \end{cases}$$

Proof.

$$\begin{aligned} p_X(x) &= \int_{\mathbb{R}} \chi_{\{0 \leq x \leq 1\}} \chi_{\{0 \leq y \leq 1\}} (x + y) \, dy \\ &= \chi_{\{0 \leq x \leq 1\}} \int_0^1 (x + y) \, dy \\ &= \chi_{\{0 \leq x \leq 1\}} \left(x + \frac{1}{2} \right) \end{aligned} \tag{10.5}$$

$$\begin{aligned} p_Y(y) &= \int_{\mathbb{R}} \chi_{\{0 \leq y \leq 1\}} \chi_{\{0 \leq x \leq 1\}} (x + y) \, dx \\ &= \chi_{\{0 \leq y \leq 1\}} \int_0^1 (x + y) \, dx \\ &= \chi_{\{0 \leq y \leq 1\}} \left(y + \frac{1}{2} \right) \end{aligned} \tag{10.6}$$

$$\begin{aligned} g_X(x) &= \int_{\mathbb{R}} \chi_{\{0 \leq x \leq 1\}} \chi_{\{0 \leq y \leq 1\}} \left(\frac{1}{2} + x \right) \left(\frac{1}{2} + y \right) \, dy \\ &= \chi_{\{0 \leq x \leq 1\}} \left(x + \frac{1}{2} \right) \int_0^1 \left(y + \frac{1}{2} \right) \, dy \\ &= \chi_{\{0 \leq x \leq 1\}} \left(x + \frac{1}{2} \right) \end{aligned} \tag{10.7}$$

$$\begin{aligned} g_Y(y) &= \int_{\mathbb{R}} \chi_{\{0 \leq x \leq 1\}} \chi_{\{0 \leq y \leq 1\}} \left(\frac{1}{2} + x\right) \left(\frac{1}{2} + y\right) dx \\ &= \chi_{\{0 \leq y \leq 1\}} \left(y + \frac{1}{2}\right) \int_0^1 \left(x + \frac{1}{2}\right) dx \\ &= \chi_{\{0 \leq y \leq 1\}} \left(y + \frac{1}{2}\right) \end{aligned} \tag{10.8}$$

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因此

$$p_X = g_X, \quad p_Y = g_Y$$